MSE 235: 2nd Year Materials Engineering Laboratory Spring 2026

Suggested Textbook / Required Course Materials

- Please have Callister on hand as a useful reference.
- A lab notebook is highly recommended to take down notes and observations about experiments.
- All students **must purchase** and wear their own protective eyewear; either goggles or glasses with side shields.

Learning Outcomes

The primary goal of this course is to provide a broad overview of the experimental, computational, mathematical, and communication tools that are used by materials engineers to characterize both a material's structure and its resulting properties. Content discussed in MSE 235 (and MSE 230) will be built upon in future MSE courses. At the completion of this course, students will be able to:

1. Use direct and indirect approaches for assessing structural and microstructural features and related properties in crystalline and non-crystalline materials.
2. Process information pertaining to atomic and molecular scale structure, microstructure, and materials properties from raw data or literature values.
3. Accurately and clearly represent engineering data in graphs, figures, and tables; analyze data and identify outlying data points using proper statistical methods.
4. Organize and write technical reports to compile and effectively describe engineering data sets and discuss related background and supporting information.
5. Identify and properly utilize primary sources and online databases to find information about material structures, properties, and processing techniques.

General Policies

Students are responsible to know the information contained in this syllabus. It is expected that students attend and actively participate in all course lectures and lab activities. All individuals will treat each other with respect and courtesy at all times. Students should arrive on time to each lecture and lab activity and refrain from leaving early, in order to minimize disruption to the course instructors, teaching assistants, and other students in the course. Additionally, students should minimize disruptive behavior while in lecture and lab, such as excessive talking, laughing, texting, and web-surfing. All course announcements will be made in lecture and/or lab and posted as "Announcements" on Brightspace. Lastly, the course instructors and TAs retain the right to ask disruptive or disrespectful students to leave the classroom or lab at any point.

Personal Protective Equipment (PPE) / Lab Safety

Labs will often involve safety hazards such as heat or chemicals. Required PPE is:

- Safety glasses
- Long hair tied back, close-toed shoes
- Any other PPE required for specific lab spaces, which will be outlined on each lab handout for the given week

Each of the labs involves special safety concerns, which will be discussed before the labs begin. If you feel any situation is unsafe, please stop what you are doing and seek the help of one of the

TAs or course instructors. Lab spaces should be returned to original/clean condition; "Leave no trace" whenever possible. The TAs should include lab cleanup as part of the students' tasks prior to being dismissed from the lab. It sets a bad example for future independent work if the students make a mess and the TA cleans it up afterwards.

*** Students not wearing the requisite PPE will be asked to do so; if the student refuses, the instructors/TAs and all other students in the course must leave the classroom/laboratory immediately. The course instructors and TAs retain the right to ask disruptive or disrespectful students to leave the classroom or lab at any point.

Assessments

Short Lab Reports (6 total)	30%
Technical Reports (2 total)	30%
Assignments, Quizzes	20%
In Lab Performance (*e.g.*, data sharing, participation, attendance)	20%

Lab reports – Written reports presenting the purpose, techniques, results, interpretation/analysis, and conclusions, from a given lab activity. Each experiment may have different required components so please read the guidelines in the lab handouts and lab templates. These will be typically due about a week after the conclusion of the lab activity. Everyone must turn in their own work.

Quizzes & Assignments – There are a few small quizzes and/or assignments related to the lab activities throughout the semester. Everyone must do their own work.

In lab performance – Students will be assessed based on attendance, sharing data with other group mates and/or classmates, and active participation in lab.

Grading Scale

Final grades will be assigned using the following scale, along with +/-. The instructors reserve the right to make slight alterations in the scale based on overall class performance. Students are responsible for checking the accuracy of the gradebook (which will be maintained on Brightspace). Please notify the instructors if you believe there are discrepancies or missing grades.

A	90	D	60
B	80	F	<60
C	70		

Deadlines

There will be **no deadline extensions or make-ups** for any graded assessment, **except for excused absences as defined by the Dean of Students** (https://www.purdue.edu/advocacy/students/absences.html). Students with excused absences will be allowed to make up missed in-class assignment and/or quizzes and will be given deadline extensions for assignments and lab reports. Without an excused absence, late assignments or lab reports will be deducted 15% of the total (original) points if submitted within 24 hrs of the posted deadline, with additional 5% deductions for each additional 24 hr period.

In case of emergency, email the instructors or call the School of Materials Engineering as soon as possible at (765) 494-4100.

Attendance
Attendance at all lectures and labs is required. No absence is permitted unless excused by the course instructors **prior to the start** of the lecture or lab session. An unexcused absence from a lab session will result in a point deduction from the "in lab performance" grade category. If you have questions about absences, contact the Dean of Students Office at odos@purdue.edu or (765) 494-1747.

Protect Purdue Guidelines
Please do not come to lab if you feel sick to prevent the spread of illnesses. Inform both instructors as soon as possible and do not come to in-person lectures or labs until you feel you have recovered. These will be considered excused absences.

Academic Integrity
"As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue."

Purdue University Regulations, Part 5, Section III- B-2-a describes the formal policies governing academic dishonesty. A guide providing specific examples, tips, and consequences is available at: http://www.purdue.edu/odos/osrr/academicintegritybrochure.php . These rules cover all graded assignments, including lab reports, quizzes, and exams. Copying or sharing any part of the graded assignment from any source is a clear violation of university policy. Transcribing, paraphrasing or following the outline of someone else's solution(s) is cheating. The test is simple: If you write down your solution after reading or hearing someone else's solution, you are cheating. Likewise, if you share or exchange your solutions with another student, in written or spoken form, you are cheating. Discussion between students of the broad concepts and general approach to lab reports is encouraged, and images, data, and results will be shared between groups; however, the solutions, charts, and graphs you turn in for grading must be your own original work. Thus, sharing work beyond raw data and broad discussions will be considered plagiarism.

Students found to engage in academic dishonesty are subject to sanctions. For a first offense, the student will receive a zero score for the assignment, report, quiz, or exam. Depending on the severity of the offense, the case may be referred to the Office of the Dean of Students for further action. A second offense will result in a failing grade for the course and definite referral to the Office of the Dean of Students. Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breeches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information that is submitted provides the greatest opportunity for the university to investigate the concern.

Use of Artificial Intelligence (AI)
All submitted assignments in MSE 235 must be the work of the student. Students are welcome to use AI (e.g., Gemini, ChatGPT, Microsoft Copilot, etc.) to brainstorm ideas to get started with

writing (i.e., get over "writer's block"). AI may also be used to improve clarity, spelling, and/or grammar of an initial draft. If the student chooses to use AI for the assignment, it must be stated in the document, preferably in the form of a reference. If AI is used for clarity, grammar, or spelling of the reports, students should also include a link to the AI "conversation" to show the initial draft. Students are also advised to use caution when using AI in general since it pulls information from across the internet, which can lead to inaccurate information.

Emergencies

In the event of a major campus emergency, course requirements, deadlines, and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances. Information about changes in this course will be posted on Brightspace and/or distributed via email to your @purdue.edu email address. Emergency preparedness is your personal responsibility. Purdue University is actively preparing for natural disasters or human-caused incidents with the ultimate goal of maintaining a safe and secure campus. For full details, visit: http://www.purdue.edu/ehps/emergency_preparedness/

- For any emergency, call 911.
- There are nearly 300 Emergency Telephone Systems throughout campus that connect directly to the Purdue Police Department (PUPD). If you feel threatened or need help, push the button and you will be connected to the PUPD.
- If we hear a fire alarm, we will immediately evacuate the building and proceed outside. Do not use the elevator.
- If we are notified of a Shelter in Place requirement for a tornado warning, we will shelter in the lowest level of this building away from windows and doors.
- If we are notified of a Shelter in Place requirement for a hazardous materials release or civil disturbance, we will shelter in our classroom shutting any open doors and windows.
- ARMS Building Emergency Plan: https://www.purdue.edu/ehps/emergency_preparedness/bep/arms- bep.html

Use of Copyrighted Materials

Among the materials that may be protected by copyright law are the lectures, notes, and other material presented in class or as part of the course. Always assume the materials presented by an instructors are protected by copyright unless the instructors have stated otherwise. Students enrolled in, and authorized visitors to, Purdue University courses are permitted to take notes, which they may use for individual/group study or for other non-commercial purposes reasonably arising from enrollment in the course or the University generally. Notes taken in class are, however, generally considered to be "derivative works" of the instructor's presentations and materials, and they are thus subject to the instructor's copyright in such presentations and materials. No individual is permitted to sell or otherwise barter notes, either to other students or to any commercial concern, for a course without the express written permission of the course instructors.

Mental Health & Wellness

Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at (765)494-6995 and http://www.purdue.edu/caps/ during and after hours, on weekends

and holidays, or through its counselors physically located in the Purdue University Student Health Center (PUSH) during business hours.

Students with Disabilities
Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: **drc@purdue.edu** or by phone: 765-494-1247.